Project Milestone 3: Initial Results and Code

Sales and Profit Analysis for Retail Decision Support Using the Global Superstore Dataset

- Sahanabanu Shaikh
- Student number - **501342966**
- Supervisor's name - Tamer Abdou
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1. Data Analysis

Dataset Overview

This project uses the Global Superstore dataset, which contains transaction-level retail data from a global company. Each row represents a single product sale and includes details such as sales amount, profit, discount, quantity, and product category. The dataset consists of 51,290 transactions and 27 variables, including both numerical and categorical features. For this analysis, the most important variables are:

- **Sales** - revenue generated from each transaction
- **Profit** - profit or loss from the transaction
- **Discount** - discount applied
- **Quantity** - number of units sold
- **Category and Sub-Category** - product classification

These variables help in understanding what drives profitability in retail transactions.

Descriptive Statistics

Initial exploration shows that:

- Sales values vary widely, indicating both small and large transactions.
- Profit includes both positive and negative values, meaning some sales result in losses.
- Discounts are not applied uniformly and vary across transactions.

The dataset contains no missing values, which makes it clean and reliable for modeling without major preprocessing.

Data Visualization and Patterns

Several visualizations were developed to explore relationships within the dataset:

- **Profit Distribution:**
  Most transactions generate small to moderate profits, but there are also noticeable losses and high-profit outliers.

- **Sales vs Profit:**
  Higher sales do not always lead to higher profit, suggesting that other factors like discounts or product types influence outcomes.
- **Discount vs Profit:**
  A clear negative relationship is observed. As discounts increase, profit tends to decrease. This is one of the most important findings in the analysis.
- **Profit by Category:**
  Profitability varies across product categories, with Technology generally performing better than Furniture and Office Supplies.
- **Profit by Region:**
  Profit also differs across regions, indicating that geographic factors may influence performance.
- **Correlation Heatmap:**
  The heatmap confirms that discount has a negative relationship with profit, while other variables show weaker correlations.

Key Insights

The exploratory analysis reveals the following:

1. Higher discounts are often linked to lower profit.
2. Profitability varies across product categories and regions.
3. There is significant variability in profit across transactions.

These observations justify the use of predictive models to support pricing and discount decisions.

2. Data Preparation

Data Cleaning

The dataset was reviewed for missing values and duplicates. Only minimal missing values were found, and duplicate records were removed to ensure data consistency.

Feature Selection

For initial modeling, the following variables were selected:

- **Input features:** Sales, Discount, Quantity
- **Regression target:** Profit

These features were chosen due to their direct relationship with transaction performance.

Data Transformation

A binary classification variable, Profit_Label, was created to support classification modeling:

- Profit > 0 = Profitable (1)
- Profit ≤ 0 = Not Profitable (0)

This transformation allows profitability to be analyzed as a classification problem.

Train-Test Split

The dataset was split into training and testing sets using an 80/20 ratio. The training data was used to build the models, while the testing data was used to evaluate performance on unseen data.

3. Model Evaluation

Regression Models

1. Linear Regression:

Linear Regression was implemented as a baseline model to predict profit. Model performance was evaluated using:

- Root Mean Squared Error (RMSE): Measures prediction error
- R^2 Score: Indicates explained variance

The model achieved:

- **RMSE:** 156.32
- **R^2:** 0.17

This performance indicates that the model explains only a small portion of the variation in profit. It suggests that the relationship between variables is not purely linear.

2. Decision Tree Regression:

A Decision Tree model was used to capture nonlinear relationships between features and profit.

The model achieved:

- **RMSE:** 123.59
- **R^2:** 0.48

This model performs significantly better than Linear Regression, indicating that nonlinear patterns play an important role in determining profit.

Classification Model

Logistic Regression:

A Logistic Regression model was developed to classify transactions as profitable or not. The model achieved good accuracy and was able to correctly identify a large proportion of profitable transactions.

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

The model achieved satisfactory performance, successfully identifying a majority of profitable and loss-making transactions.

Overall Findings

The results of the initial models indicate:

- The Decision Tree model outperforms Linear Regression, showing that profit relationships are nonlinear.
- Logistic Regression can effectively identify profitable versus loss-making transactions.
- Discount is a key factor influencing profitability.

Overall, these results demonstrate that predictive models can be useful tools for improving retail decision-making.

4. Code Documentation

All analysis and modeling were implemented in Python using Jupyter Notebook.

The project is organized within a GitHub repository, which includes:

- Jupyter Notebook containing data analysis and models
- Dataset used for the project
- PDF version of the report
- README file describing the project structure and stages

GitHub Repository:
https://github.com/sahanashaikh14/global-superstore-profit-analysis

5. Video Presentation

A five-minute video titled **"Initial Results and Code"** was created using Zoom. The video has been uploaded and is accessible through the GitHub repository as part of the project submission.

https://github.com/sahanashaikh14/global-superstore-profit-analysis/blob/98929208ab505085a9827a924e23af7453cf7756/Milestone%203%20Video%20Presentation.mp4

The presentation includes:

- Overview of the dataset and objectives
- Key findings from exploratory data analysis
- Explanation of preprocessing steps
- Walkthrough of regression and classification models

- Summary of results and insights

6. References

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